EXPERIMENT : No. 1 Practical No. 1

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Date

Aim : Creation of Test Suite using Selenium IDE Tool : Install Selenium IDE (chrome / firefox Extension) and create a test suit containing a minimum of four test cases to validate different web page format (HTML page, XML response, JSON API output and a dynamic from page).
Execute the suit and export the automation script in WebDriver format.

Requirements :
1) Selenium IDE extension on a browser.
2) NetBeans IDE for creating test case files.

Steps -
1. Open Google and search "Selenium IDE chrome."
2. Open the first result and click "Add to chrome" to install Selenium IDE.
3. For testing, we need a test case suite, so we create a WebApplication in NetBeans IDE.
File → New Project → JavaWeb → Web Applicatio → Add GlassFish Server → Finish.
4. Open index.html and write the code for the "Arithmetic Operations" test case suite.

Sundaram

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index.html

```
<html>
 <head>
  <title> prac1 </title>
  <script language = "javascript">
   function addition()
    {
     var num1 = parseInt(document.arithmetic.n1.value);
     var num2 = parseInt(document.arithmetic.n2.value);
     var result = num1 + num2;
     document.arithmetic.res.value = result;
    }
   function subtraction()
    {
     var num1 = parseInt(document.arithmetic.n1.value);
     var num2 = parseInt(document.arithmetic.n2.value);
     var result = num1 - num2;
     document.arithmetic.res.value = result;
    }
   function multiplication()
    {
     var num1 = parseInt(document.arithmetic.n1.value);
     var num2 = parseInt(document.arithmetic.n2.value);
     var result = num1 * num2;
     document.arithmetic.res.value = result;
    }
   function division()
    {
```

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```
var num1=parseInt(document.arithmetic.n1.value);
var num2=parseInt(document.arithmetic.n2.value);
var result=num1/num2;
document.arithmetic.res.value=result;
}
</script>
</head>
<body>
<h1 align="center"> Arithmetic Operations </h1>
<form name="arithmetic">
<table border="1" align="center">
<tr>
<td> Number1 : </td>
<td><input type="text" name="n1" size="20"> </td
</tr>
<tr>
<td> Number 2: </td>
<td><input type="text" name="n2" size="20"> </td
</tr>
<tr>
<td colspan=2>
<input type="button" name="add"
value="Add" onclick="javascript:addition();">  
<input type="button" name="sub"
value="Subtract" onclick="javaseript:subtraction();">  
<input type="button" name="mul"
value="Multiply" onclick="javascript:multiplication();">  
```


Teacher's Sign. :

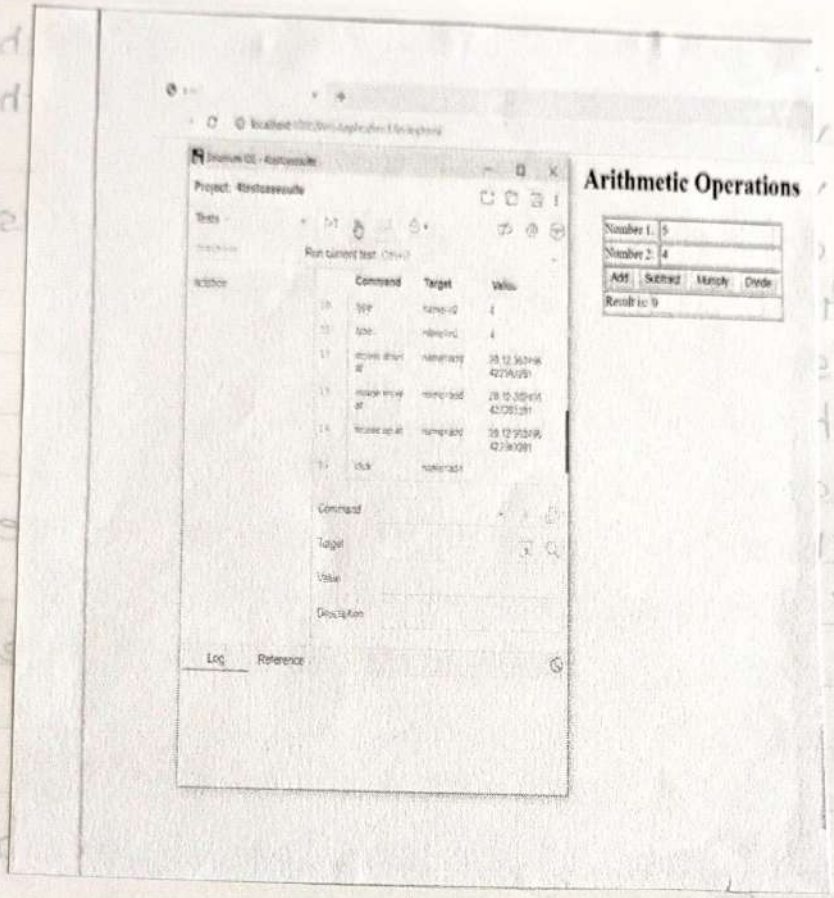
```
<input type="button" name="div" value="Divide"
onclick="javascript:division();">  
</td>
</tr>
<tr>
<td colspan="2"> Result is: <input type="text"
name="res" size="20" value=" "></td>
</tr>
</table>
</form>
</body>
</html>
```

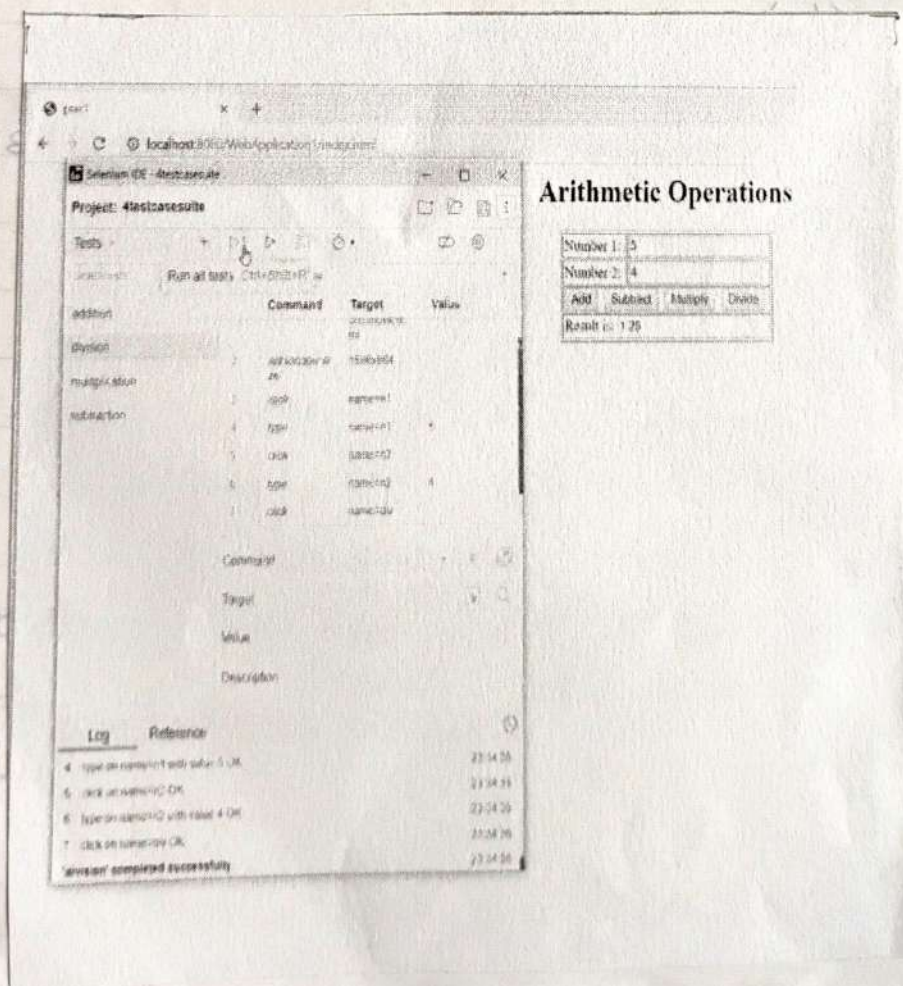
5. Create a JSP File "newjsp.jsp" with a link to the created HTML File.

newjsp.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html;
charset=UTF-8">
<title> JSP Page </title>
</head>
<body align="center">
<h1> Test Cases IDE </h1>
<a href="index.html"> Arithmetic Operations</a>
</body>
</html>
```

output:


Arithmetic Operations
Number 1: 5
Number 2: 4
Add
Subtract
Multiply
Divide
Result is: 9


Arithmetic Operations
Number 1: 5
Number 2: 4
Add
Subtract
Multiply
Divide

EXPERIMENT : No. Page No. Date

6. Run newjsp.jsp from NetBeans. it will open in a browser.
- Open the ARITHMETIC OPERATIONS page using the given link.
- Copy the page URL and open selenium IDE.
- Create a New project and enter the project name.
- Paste the URL and start Recording.
- A new tab will be opened, maximize that tab (this tab is your working tab for recording)
- Perform the required actions on the Arithmetic Operation

7. Click on stop Recording:
- Give the test case a name, such as "addition".
- Save the project. It will be saved with a .side extension.
- Run the addition test and check the result in the Log area in the Selenium IDE.

Note 1: You can set the "Test Execution speed" too.
Note 2: Paste the URL again if the GlassFish Error occurs between any test case.

8. Now create 3 more test cases (subtraction, multiplication, division). Then "Run all tests".

9. Finish!

Teacher's Sign. :

EXPERIMENT : No. 2 Practical No. 2 Page No. Date

Aim : Cross - Website Functional Testing Using Selenium IDE Tool : Selenium IDE Develop and execute a test suite for two different websites.
Perform functional operations such as clicking links, filling forms, selecting dropdown options and verifying displayed content. Analyze test results and generate execution logs.

Requirements : 1) Selenium IDE extension on a browser
2) A stable internet connection.

Steps :

1) Search Google for " Selenium IDE chrome". Open the first relevant result and add the " Selenium IDE" extension to the browser.

2) For testing, use redbus.in as the first website. we'll be checking some selenium IDE commands like assertText, verify Title, storeText and echo.

3) Open redbus.in in the browser. Copy the URL of the page and then open selenium IDE extension.

4) In Selenium IDE, choose the option to record a new test project in a new project.
- Enter a suitable project name and click OK.
- In the " Set your project's base URL" box, paste the RedBus URL you copied.
- Click "Start Recording".

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- A new tab will be opened, maximize that tab (This tab is your working tab).

5. In the working redbus.in tab, right-click anywhere on the webpage. Select Selenium IDE and click Verify Title.
   - Check the verifyTitle command is recorded in Selenium IDE.
   - Fill in the FROM location, TO location and Onward Date fields on redbus.in
   - Click "Search Buses" to continue the test. (Avoid unnecessary clicks)

6. After the bus search results appear, find the bus provider name.
   - Right-Click it → Selenium IDE → Assert Text

7. Find the ticket price on the page.
   - Right-click the price → Selenium IDE → store Text to save the price for later use.
   - For storing the Price, Selenium IDE prompts you to enter the name of the variable.
   - Enter the variable name "price" and click OK.
   - Confirm that Selenium IDE automatically maps the selected go page element to its css target.

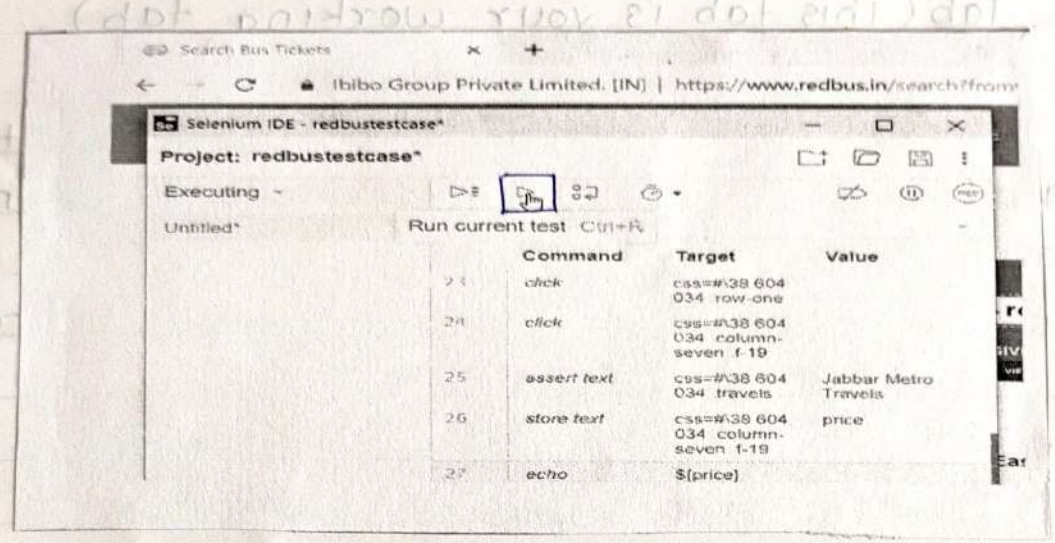
EXPERIMENT : No.

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Date

8. Click on the next empty line in Selenium IDE. Enter the echo command with the variable ${price} to display the stored price.

9. Now we'll Run current test.
   - After the test is completed, check the log area in Selenium IDE.
   - You will see "echo: 350" as the output. This means 350 was stored in the variable "price" using storeText.

10. Finish!


Search Bus Tickets
Ibibo Group Private Limited. [IN] | https://www.redbus.in/search?from
Selenium IDE - redbustestcase*
Project: redbustestcase*
Executing
Untitled*
Run current test Ctrl+R
Command
Target
Value
click
css=#\38 604 034 row-one
click
css=#\38 604 034 column-seven f-19
assert text
css=#\38 604 034 travels
Jabbar Metro Travels
store text
css=#\38 604 034 column-seven f-19
price
echo
${price}

Output :


redDeals
SORT BY
15:15
20:00
07:00
11:30
IMPORTANT TIP

EXPERIMENT : No. 3 Practical No. 3

Page No. Date

Aim : Object Identification and Counting on Web Page
Tool : Selenium WebDriver.
Write a program to identify and count the total number of objects present on a web page including links, images, buttons, and input elements.
Display categorized counts.

Pre - Requisites :
1> check that you have JDK.
2> check that you have Eclipse IDE.
3> check that you have Selenium Server Driver and Client Driver (JAR Files).
4> check that you have Gecko Driver.
5> check that you have a stable Internet connection.

Steps :
1. Open Eclipse. Select your workspace directory. Click Launch.

2. Create a Project (File > New > Java Project)

3. Name the project as "p3" > click Finish > click Don't create module.

4. Close the "Welcome" tab and Create a package (right-click on Project Name > New > package > Name it > Finish).

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5. Create a Class (right-click on Project Name > New > class > Name it > Finish).

6. For Adding "Selenium Server Driver and Client Driver (JAR Files)" in Eclipse IDE :
   - right-click on project name > Build Path > Configure Build Path....
   - Select Java Build Path > Libraries > Classpath > click Add External JARs..
   - Browse to the JAR files Then click Apply and close.

7. Create the Java Script (This script will be run by Eclipse IDE)
   (This script tells Eclipse to open FireFox, visit Google, find the webpage elements with the <a> tag using Selenium to show the result).

8. Then put the path of "geckodriver" in a String driverPath.

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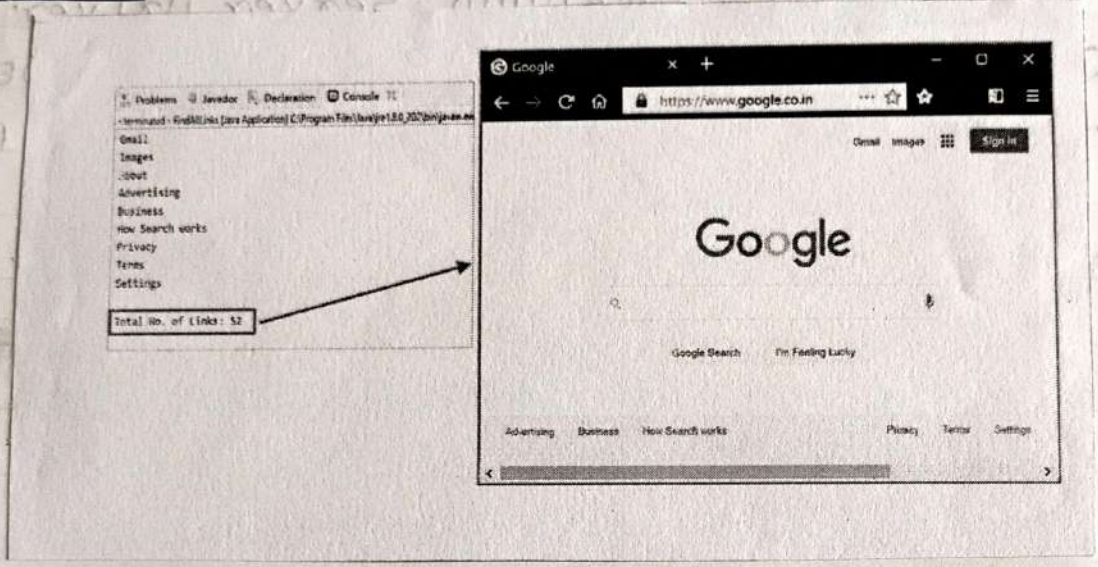
— FindAllLinks.java —

```
Package p3;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;

public class FindAllLinks {
   static String driverPath = "C:\Users\Student\STQA \
     geckodriver.exe";
   public static void main (String[] args) {
        System.setProperty ("webdriver.gecko.driver",
            driverPath);
        WebDriver driver = new FirefoxDriver ();
        String appUrl = "https://www.google.co.in/";
        driver.get (appUrl);
        java.util.List <WebElement> links = driver. -
FindElements (By.tagName ("a"));
        for (int i=1; i<links.size(); i= i+1)
        {
        System.out.println(links.get(i).getText());
        }
        System.out.println("Total No. of Links: "+ links.size());
        }}
```

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output:


Problems
Javadoc
Declaration
Console
Gmail
Images
About
Advertising
Business
How Search works
Privacy
Terms
Settings
Total No. of links: 52
Google
https://www.google.co.in
Gmail
Images
Sign in
Google
Google Search
I'm Feeling Lucky
Advertising
Business
How Search works
Privacy
Terms
Settings

EXPERIMENT : No. 4. Practical No. 4

Page No. Date

Aim : login Automation and Validation
Tool : Selenium WebDriver (Java / PHP)
Write an automation script to perform login on a specified web page. Verify successful login using assertions such as URL validation, welcome message validation, and logout visibility check.

Pre-Requisites :
1) check that you have JDK
2) check that you have Eclipse IDE.
3) check that you have Selenium Server Driver and client Driver (JAR files).
4) check that you have Gecko Driver.
5) check that you have a stable internet connection.

Steps :
1. Open Eclipse IDE and select the workspace. Then Go to File → New → Java Project.

2. Name the project as "p4" and click Finish.

3. Create a package : Right-click project → New → Package → Enter name → Finish.

4. Create a class : Right-click project → New → class → Enter name → Finish.

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5. Add Selenium JAR files :
Right-click project → Build Path → configure Build Path → Libraries → Add External JARs → Select files → Apply and close.

6. Open any browser and go to the facebook login page.
- Find the Email textbox → Right-click on the Email textbox → click Inspect.
- The browser developer tools will open.
- Find the HTML code of the selected Email textbox.
- Look for the id attribute and note its value.

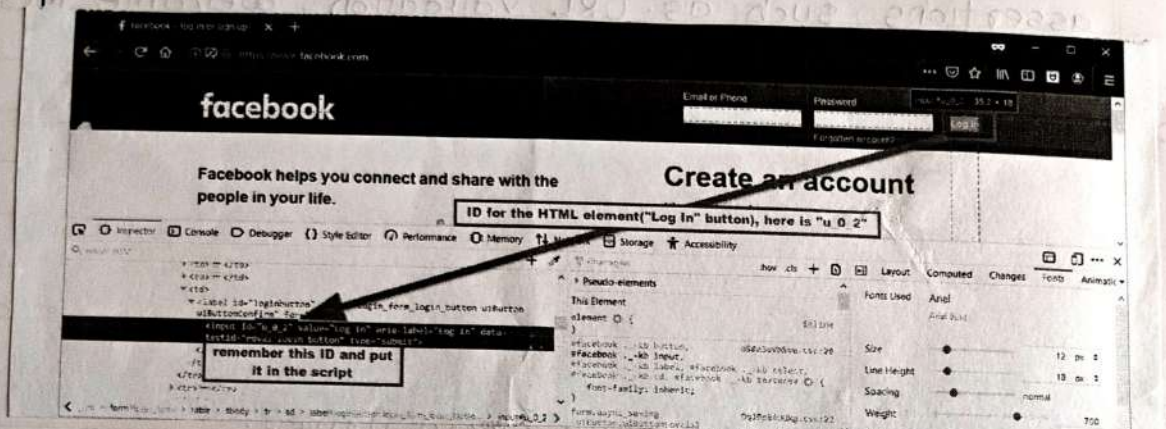
(Do same for Login Button and Password textbox)

7. Write the Selenium Java program. Set the GeckoDriver path and put the ID's in the .id() method respectively.

— FB_login.java —

```
package p4;
import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
```

step 6-


facebook
Email or Phone
Password
Log In
Facebook helps you connect and share with the people in your life.
Create an account
ID for the HTML element("Log In" button), here is "u_0_2"
remember this ID and put it in the script
Inspector
Console
Debugger
Style Editor
Performance
Memory
Storage
Accessibility
Layout
Computed
Changes
Fonts
Animations
Pseudo-elements
This Element
Fonts Used
Arial
Size
Line Height
Spacing
Weight
normal
700

EXPERIMENT : No.

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```
import org.openqa.selenium.firefox.internal.ProfilesIni;

public class FB_login {
   static string driverPath = "C:\\Users\\Usman\\503\\
                              geckodriver.exe";
   public static void main (string[] args) {
    System.setProperty ("webdriver.gecko.driver", driverPath);
     ProfilesIni allProfiles = new ProfilesIni();
     FirefoxProfile fp = new FirefoxProfile();

     fp.setPreference (FirefoxProfile.PORT_PREFERENCE, "7055");
     FirefoxOptions options = new FirefoxOptions();
     options.setProfile (fp);

WebDriver driver = new FirefoxDriver (options);
String appUrl = "https://www.facebook.com/";
driver.get (appUrl);
driver.manage().window().maximize();

String expectedTitle = "Facebook - log in or sign up";
String actualTitle = driver.getTitle();
if (expectedTitle.equals (actualTitle)) {
      System.out.println ("Verification Successful -
The correct title is displayed on the web page."); }
else {
      System.out.println ("Verification failed - An
incorrect title is displayed on the web page.");
    }
```

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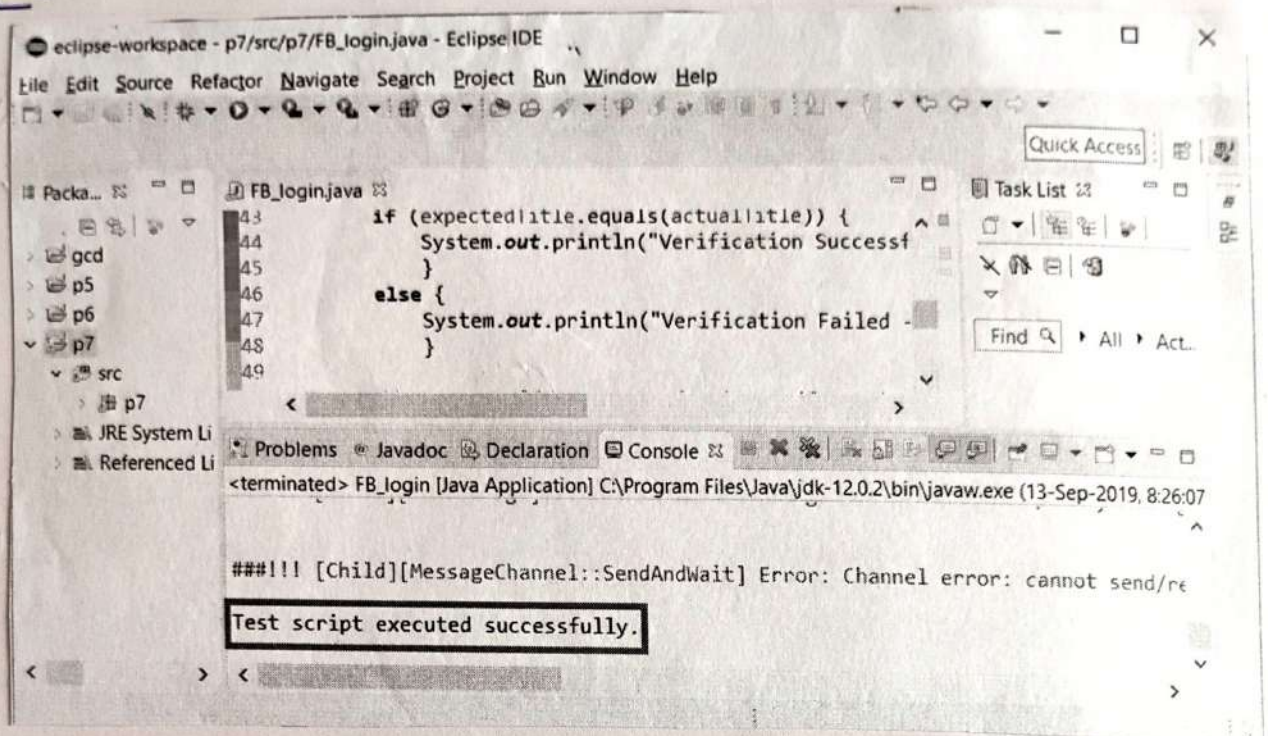
```
WebElement username = driver.findElement(By.id
                                ("email"));
username.clear();
username.sendKeys("your email id");

WebElement password = driver.findElement(By.id("pass"));
password.clear();
password.sendKeys("your password");

password.sendKeys(Keys.ENTER);
WebElement LogInButton = driver.findElement(By.id
                                ("u_0_2"));
LogInButton.click();
driver.close();
System.out.println("Test script executed successfully");
System.exit(0);
}
}
```

8. Run the Java file from Eclipse IDE and check the output.

output -


eclipse-workspace - p7/src/p7/FB_login.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Quick Access
Packa...
gcd
p5
p6
p7
src
p7
JRE System Li
Referenced Li
FB_login.java
43 if (expectedTitle.equals(actualTitle)) {
44 System.out.println("Verification Successf
45 }
46 else {
47 System.out.println("Verification Failed -
48 }
49
Task List
Find
All
Act...
Problems
Javadoc
Declaration
Console
<terminated> FB_login [Java Application] C:\Program Files\Java\jdk-12.0.2\bin\javaw.exe (13-Sep-2019, 8:26:07
###!!! [Child][MessageChannel::SendAndWait] Error: Channel error: cannot send/re
Test script executed successfully.

EXPERIMENT : No. 5 Practical No. 5

Page No.
Date

Aim : Selenium Server (Grid) Configuration and Remote Execution.

Tool : Selenium Server (Grid). Install and configure selenium server in Hub-Node architecture. Execute a test script on multiple browsers using Remote WebDriver and compare cross-browser results.

Requirements :

Java JDK 11 or above
Selenium server (latest JAR)
Chrome browser
Firefox browser
ChromeDriver and GeckoDriver (optional for older versions)
Eclipse / IntelliJ IDEA
Selenium Java library

Steps :

1. Install Java JDK 11 or above.

2. Download the selenium server JAR from the official selenium website.

3. Create a Selenium folder in C Drive and place the JAR file in it.
   - Open Command Prompt
   cd C:\selenium
   C:\selenium>java -jar selenium-server-4.46.0-
   jar info.

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C:\selenium > java -jar selenium-server-4.46.0.jar hub

4. Start the Selenium Hub using the Selenium Server command - http://localhost:4444

5. Open another Command Prompt and start a Selenium Node.
   - Go to Selenium folder: cd C:\Selenium
   Start Node :
   java -jar selenium-server-4.46.0.jar node

6. Add Firefox Node (Optional)
   If Firefox is installed: open another command promp
   Run: java -jar selenium-server-4.46.0.jar node --browser
   Firefox

7. Open Eclipse and create a new java project.
   (Create a package and java classes in project)

8. Add the required Selenium libraries / JAR files to the project build path.

9. Write the selenium java program.
   (Create two class files - grid1.java & grid2.java)

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Page No.
Date

— grid1.java —

```
package seleniumgrid;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;

public class grid1 {
  public static void main(String[] args) throws Exception {
    ChromeOptions options = new ChromeOptions();

    WebDriver driver = new RemoteWebDriver(
      new URL("http://localhost:4444"),
      options);

      driver.get("https://www.google.com");
      System.out.println(driver.getTitle());
      driver.quit();
  }

}
```

Output - Grid 1 :-

Problems Javadoc Declaration **Console** x

```
<terminated> grid1 [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe  (11-Aug-2026, 12:26:42 pm)
Aug 11, 2026 12:26:45 PM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
google
```

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— grid2.java —

```
package seleniumgrid;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;

    public class grid2 {
     public static void main (String [] args) {
      try {
       URL gridURL = new URL ("http://localhost:4444");
       FirefoxOptions options = new FirefoxOptions();

       WebDriver driver= new RemoteWebDriver(
                                gridURL, options);
      driver.get ("https://www.google.com");
      System.out.println ("Page Title:" + driver.getTitle())

      driver.manage().window().maximize();
      Thread.sleep (5000);
       driver.quit ();
      } catch( Exception e) {
        e.printStackTrace ();
        }
    }
    }
```

10. Run the Java file from Eclipse IDE.

output : Grid 2-

Problems Javadoc Declaration **Console** x

```
<terminated> grid2 [Java Application] C:\Program Files\Java\jdk-17\bin\javaw.exe  (11-Aug-2026, 12:27:02 pm)
Aug 11, 2026 12:27:06 PM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Page Title: Google
```

EXPERIMENT : No. 6 Practical No. 6

Page No.
Date

Aim : Cross - Browser Testing Using Selenium Grid Tool: Selenium Grid Execute automation script across multiple browser (chrome, Firefox, Edge) using Selenium Grid. Compare execution time and compatibility issues.

Requirements -
Java JDK 11 or above
Selenium server (latest JAR)
Chrome browser
Firefox browser
Edge browser
ChromeDriver and GeckoDriver
Selenium Java library

Steps -
1. Install Java JDK 11 or above and Download the selenium server JAR.

2. Create a selenium folder in C Drive and place JAR File in it
· Open command prompt -
cd C:\selenium
C:\selenium > java -jar selenium-server-4.46.0.jar info

C:\selenium > java -jar selenium-server-4.46.0.jar hub

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3. Start the selenium Hub using the selenium server command - http://localhost:4444

4. Open another cmd from selenium Folder and Start node- java -jar-selenium-server-4.46.0.jar node

5. Add Firefox node
Open another cmd from selenium folder and enter
java -jar selenium-server-4.46.0.jar node --browser-Firefox

6. Open Eclipse and create a new project.
(create package and class file in project)

7. Add required selenium JAR files to the project build path.

8. Idrite the selenium java program -

— grid3.java —

```
package seleniumGrid1;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.edge.EdgeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
```

Sundaram

Teacher's Sign. : _____

EXPERIMENT : No.

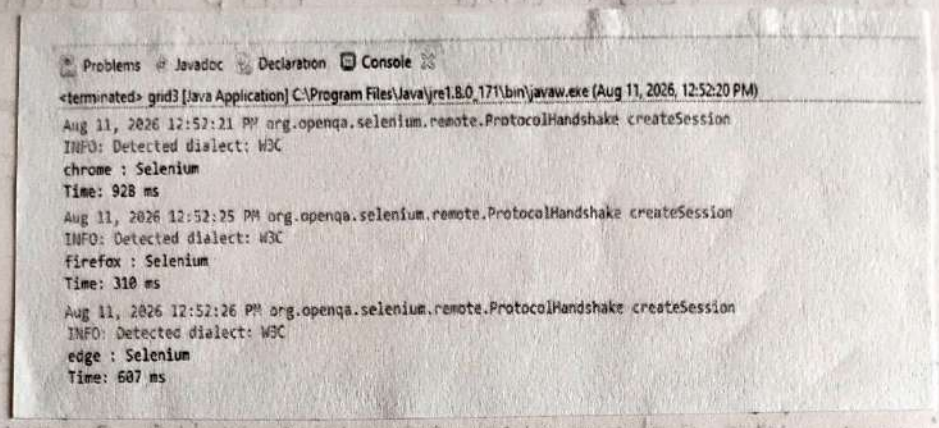
Page No.
Date

```
public class grid3 {
    public static void main(String[] args) throws Exception{
String[] browsers = {"chrome", "firefox", "edge"};
for (String browser : browsers) {
  WebDriver driver;
  if (browser.equals("chrome"))
   driver = new RemoteWebDriver(
     new URL("http://localhost:4444"),
     new ChromeOptions());
  else if (browser.equals("firefox"))
    driver = new RemoteWebDriver(
     new URL("http://localhost:4444"),
     new EdgeOptions());
  long start = System.currentTimeMillis();
  driver.get("https://www.selenium.dev/");
  System.out.println(browser+": "+driver.getTitle());
  long time=System.currentTimeMillis() - start;
  System.out.println("Time: "+time+"ms");
  driver.quit();
  }
 }
}
```

9. Run the java file from Eclipse.

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Problems
Javadoc
Declaration
Console
<terminated> grid3 [Java Application] C:\Program Files\Java\jre1.8.0_171\bin\javaw.exe (Aug 11, 2026, 12:52:20 PM)
Aug 11, 2026 12:52:21 PM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
chrome : Selenium
Time: 928 ms
Aug 11, 2026 12:52:25 PM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
firefox : Selenium
Time: 310 ms
Aug 11, 2026 12:52:26 PM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
edge : Selenium
Time: 607 ms

EXPERIMENT : No. 7. Practical No. 7

Page No. Date

Aim : List and Combo Box item verification Tool : Selenium WebDriver Automate the process of - identifying a list or dropdown box on a web page and determine the total number of items validate default selected value and perform selection testing.

PRE - REQUISITES :

1) Check that you have JDK , Eclipse IDE , Selenium Server Driverand Client Driver (JAR files).
2) Gecko Driver and a stable Internet connection.

Steps-

1. Open Eclipse and create a java project. (create package and class file in project).

2. Add JAR files in project from build path.

3. Create HTML file in a notepad > save it > open it in browser > Copy the URL

— (combobox.html) —

```
<select id = "continents">
    <option value = " Asia "> Asia </option>
    <option value = "Europe"> Europe </option>
    <option value = "Africa"> Africa </option>
</select>
```

Sundaram

Teacher's Sign. : _____

EXPERIMENT : No. _____ Page No. Date

4. Creating the script in JAVA:
- Put the path of our LOCAL file (combobox.html) in string.
- Also put the path of "geckodriver" in a driverpath.

— (combobox.java) —

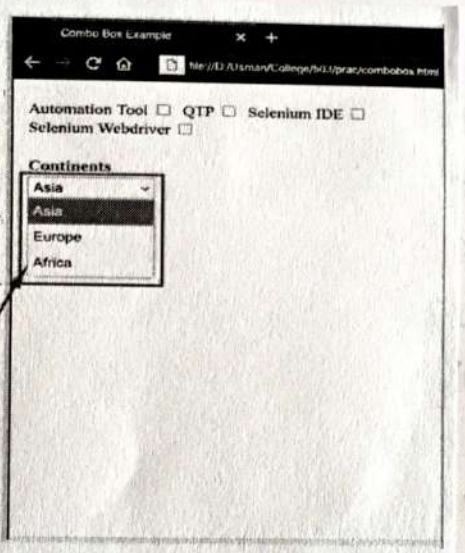
```
package p7;
import java.util.List;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;
import org.openqa.selenium.Support.ui.Select;

public class ComboBox{
    static String driverPath = "C:\\Users\\Usman\\
                    \\503\\geckodriver.exe";
    public static void main(String[] args){
    System.setProperty("webdriver.gecko.driver",
                                    driverPath);
    WebDriver driver = new FirefoxDriver();
    String appUrl = "https://www.toolsqa.com/automation
        -practice-form 1";
    driver.get(appUrl);
    Select oSelect = new
Select(driver.findElement(By.id("continents")));
```

4.

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
public class ComboBox {
    public static void main(String[] args) throws
        // Copy and save the path as we'll use it in the script
        System.setProperty("webdriver.gecko.driver",
        "D:\\Usman\\College\\503\\geckodriver.exe");
        WebDriver driver = new FirefoxDriver();
        driver.get("file:///D:/Usman/College/503/prac/
        combobox.html");
        WebElement e = driver.findElement(By.id("continents"));
        Select s = new Select(e);
        s.selectByVisibleText("Asia");
        Thread.sleep(2000);
        }
}
```

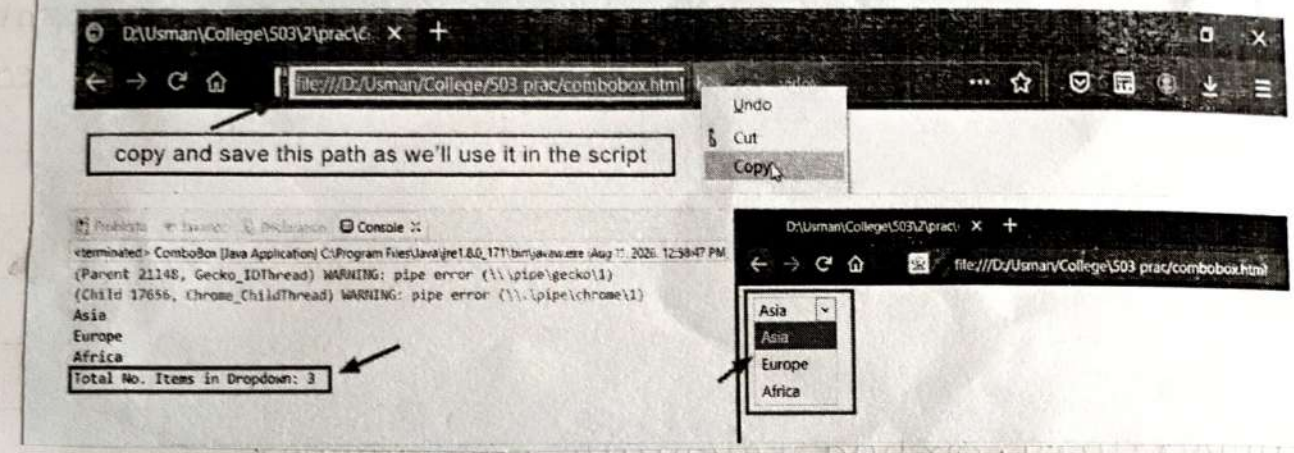
copy and save this path as we'll use it in the script



```
List <WebElement> oSize = oSelect.getOptions();
int iListSize = oSize.size();
for (int i = 0; i < iListSize; i++)
{
    String sValue = oSelect.getOptions().get(i).get-
                                                Text();
System.out.println(sValue);
}
System.out.println("Total No. Items in
                Dropdown: " +iListSize);
    }
}
```

5. Run the file from Eclipse IDE.

output -


file:///D:/Usman/College/503 prac/combobox.html
copy and save this path as we'll use it in the script
Undo
Cut
Copy
Console
<terminated> ComboBox [Java Application] C:\Program Files\Java\jre1.8.0_171\bin\javaw.exe (Aug 11, 2026, 12:58:47 PM)
(Parent 21148, Gecko_IDThread) WARNING: pipe error (\\.\pipe\gecko\1)
(Child 17656, Chrome_ChildThread) WARNING: pipe error (\\.\pipe\chrome\1)
Asia
Europe
Africa
Total No. Items in Dropdown: 3
file:///D:/Usman/College/503 prac/combobox.html
Asia
Asia
Europe
Africa

EXPERIMENT : No. 8 Practical No. 8

Page No.
Date

Aim : Checkbox Identification and Validation Tool: Selenium WebDriver Write a program to count the total number of checkboxes on a web page, including checked and unchecked counts. Perform dynamic selection and validation.

Pre - Requisities

1) Check that you have JDK, Eclipse IDE, Selenium Server driver & client driver (JAR files).
2) Gecko Driver

Steps:

1. Open Eclipse and create a java project. and creat package a class file in project.

2. Add JAR files in project.

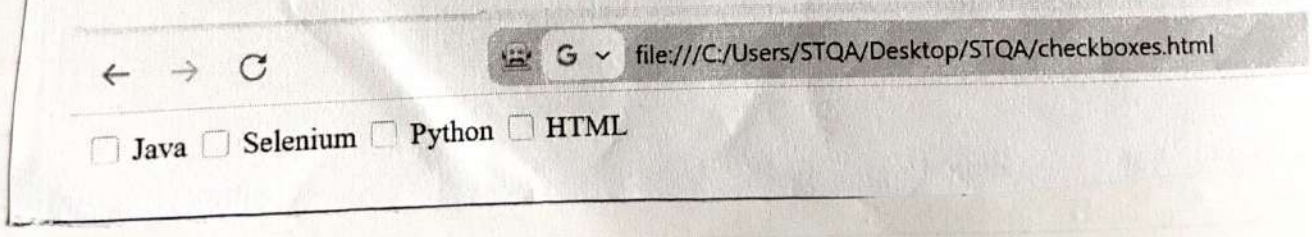
3. Create HTML file in notepad > Save it > open it in browser > copy the URL

— (checkboxes.html) —

```
<input type="checkbox"> Java
<input type = "checkbox"> Selenium
<input type = "checkbox"> Python
<input type = "checkbox"> HTML
```

Teacher's Sign. :

checkboxes.html - output



EXPERIMENT : No.

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Date

4. Create the script in java.
- put the path of your local file (checkboxes.html)
- Also put the path of "geckodriver" in a driver pat

— (checkbox.java) —

```
package studentsdata;
import org.openqa.selenium.*;
import org.openqa.selenium.firefox.FirefoxDriver;
import java.util.List;

public class checkbox {
    public static void main (String[] args) {
        System.setProperty("webdriver.gecko.driver",
            "C:\\Users\\Student\\STQA\\geckodriver.exe");

        WebDriver driver = new FirefoxDriver();
        driver.get("file://C:/Users/Desktop/STQA/checkboxes
                                                .html");
        List<WebElement> checkboxes =
            driver.findElements(By.xpath("//input[@type=
                                    'checkbox']"));
        int checked = 0;
        int unchecked = 0;
        for(WebElement checkbox : checkboxes) {
            if (cheakbox.isSelected())
                checked++;
            else
                unchecked++;
```

Teacher's Sign. :

```
}
System.out.println ("Total :" + checkboxes.size());
System.out.println ("checked:" + checked);
System.out.println ("Unchecked:" + unchecked);

for (WebElement checkbox : checkboxes) {
   if (!checkbox.isSelected()) {
        checkbox.click();
        if (checkbox.isSelected())
            system.out.println ("checkbox selected
                                 successfully");
        break;
      }
    }
    driver.quit();
  }
}
```

5. Run the file from Eclipse IDE.

output -

```
41
42          System.out.println("Total: " + checkboxes.size());
43          System.out.println("Checked: " + checked);
44          System.out.println("Unchecked: " + unchecked);
45
46          for (WebElement checkbox : checkboxes) {
```

Problems Javadoc Declaration Console

```
<terminated> p9 [Java Application] C:\Program Files\Java\jdk-21.0.12.1\bin\javaw.exe (Sep 8, 2026, 6:04:09 PM)
Total: 4
Checked: 0
Unchecked: 4
Checkbox selected successfully
```

EXPERIMENT: No. 9 Practical No. 9

Page No.
Date

Aim: Dynamic web table Handling and sorting Validation Tool: Selenium WebDriver. Automate extraction of data from a dynamic web table. Validate sorting functionality and verify correctness of filtered results.

Steps:

1. Open Eclipse and Create a project. (create package and class file in project).

2. Add JAR Files in project.

3. Write the script in java.
   - put the path of geckodriver in driverpath.

— (DynamicTableTest.java) —

```
package p9;
import java.util.*;
import org.openqa.selenium.*;
import org.openqa.selenium.firefox.FirefoxDriver;

public class DynamicTableTest {
    public static void main(String[] args) {
        System.setProperty("webdriver.gecko.driver",
        "C:\\program files\\Mozilla Firefox\\geckodriver.exe");

        WebDriver d = new FirefoxDriver();
        d.get("https://the-internet.herokuapp.com/tables");
```

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EXPERIMENT : No.

Page No.
Date

```
List <WebElement> rows =
d.findElements(By.cssSelector("#table1 tbody tr"));

for(WebElement row: rows)
  System.out.println(row.getText());
d.findElement(By.cssSelector("#table1 th:nth-child(1)")).
                                        click();
rows=d.findElements(By.cssSelector("#table1 tbody tr"));

List <String> actual = new ArrayList<>();
for(WebElement row: rows)
 actual.add(row.findElements(By.tagName("td")).
                        get(0).getText());
List <String> expected = new ArrayList<>(actual);
Collections.sort(expected);

if(actual.equals(expected))
  System.out.println("Sorting: PASS");
else
  System.out.println("Sorting: FAIL");
d.quit();
}
```

4. Run the program from Eclipse IDE.

Teacher's Sign. :

output -

```
Aug 22, 2026 10:47:42 AM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Smith John jsmith@gmail.com $50.00 http://www.jsmith.com edit delete
Bach Frank fbach@yahoo.com $51.00 http://www.frank.com edit delete
Doe Jason jdoe@hotmail.com $100.00 http://www.jdoe.com edit delete
Conway Tim tconway@earthlink.net $50.00 http://www.timconway.com edit delete
Sorting: PASS
1787375065974  Marionette      INFO    Stopped listening on port 37711
JavaScript error: resource://gre/modules/PrivateBrowsingUtils.sys.mjs, line 50: TypeError: can't access property "QueryInterface", aWindow.docShell is null
Dynamically enable window occlusion 1
```

Data Tables

Often times when you see a table it contains data which is sortable -- sometimes with actions that can be taken within each row (e.g. edit, delete). And it can be challenging to automate interaction with sets of data in a table depending on how it is constructed.

Example 1

No Class or ID attributes to signify groupings of rows and columns

Last Name	First Name	Email	Due	Web Site	Action
Bach	Frank	fbach@yahoo.com	$51.00	http://www.frank.com	edit delete
Conway	Tim	tconway@earthlink.net	$50.00	http://www.timconway.com	edit delete
Doe	Jason	jdoe@hotmail.com	$100.00	http://www.jdoe.com	edit delete
Smith	John	jsmith@gmail.com	$50.00	http://www.jsmith.com	edit delete

Example 2

Class and ID attributes to signify groupings of rows and columns

Last Name	First Name	Email	Due	Web Site	Action
Smith	John	jsmith@gmail.com	$50.00	http://www.jsmith.com	edit delete
Bach	Frank	fbach@yahoo.com	$51.00	http://www.frank.com	edit delete
Doe	Jason	jdoe@hotmail.com	$100.00	http://www.jdoe.com	edit delete
Conway	Tim	tconway@earthlink.net	$50.00	http://www.timconway.com	edit delete

Powered by Elemental Selenium

EXPERIMENT : No. 10. Practical No. 10

Page No.
Date

Aim : File upload and Download Automation Tool : Selenium WebDriver + Robot class Develop a script to automate file upload and download functionality. Validate successful upload and verify file existence after download.

Steps -

1] Open Eclipse and create a java project with package and class file.

— (FileTest.java) —

```
package practical 10;
import java.io.File;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxBinary;
import org.openqa.selenium.firefox.FirefoxDriver;

public class FileTest {
   public static void main (String[] args ) throws
Exception {
 System.setProperty("webdriver.gecko.driver",
   "C:\\Program Files\\Mozilla Firefox\\geckodriver.exe");
 FirefoxBinary firefoxBinary =
   new FirefoxBinary ( new File (
        "C:\\Program Files\\Mozilla Firefox\\firefox.exe")
   );
```

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EXPERIMENT : No.

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```
WebDriver d = new FirefoxDriver();
d.get ("https://the-internet.herokuapp.com/upload");
File uploadFile = new File ("C:\\TestFiles\\upload.txt");

d.findElement (By.id ("upload"))
  .sendKeys (uploadFile.getAbsolutePath ());

d.findElement(By.id ("submit")).click();
System.out.println ("Upload SuccessFully")
 }
}
```

step 2] Folder setup -
Create thise folder:
C:\TestFiles and creat text file in notpad.
put your file here : C:\TestFiles\upload.txt

step 3] Run your script from Edipse IDE.

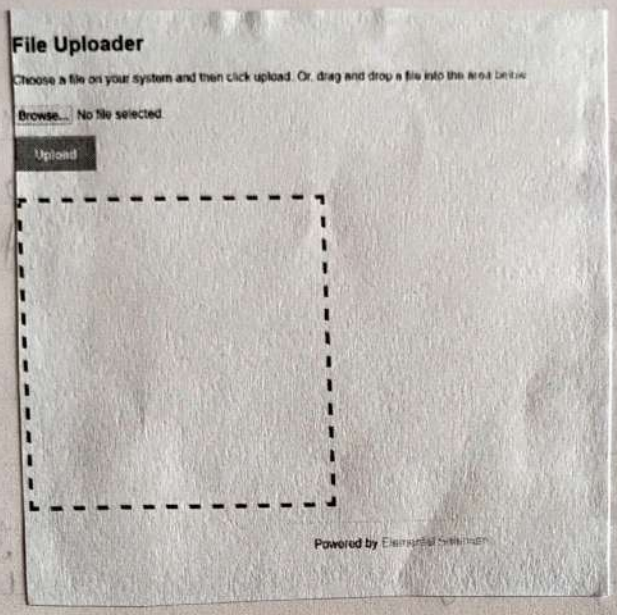
step 4] Upload your text file in File uploader.

step 5] Then Go to Url of file uploader and type download at last then enter.
you will see your uploaded text file in File Downloader.

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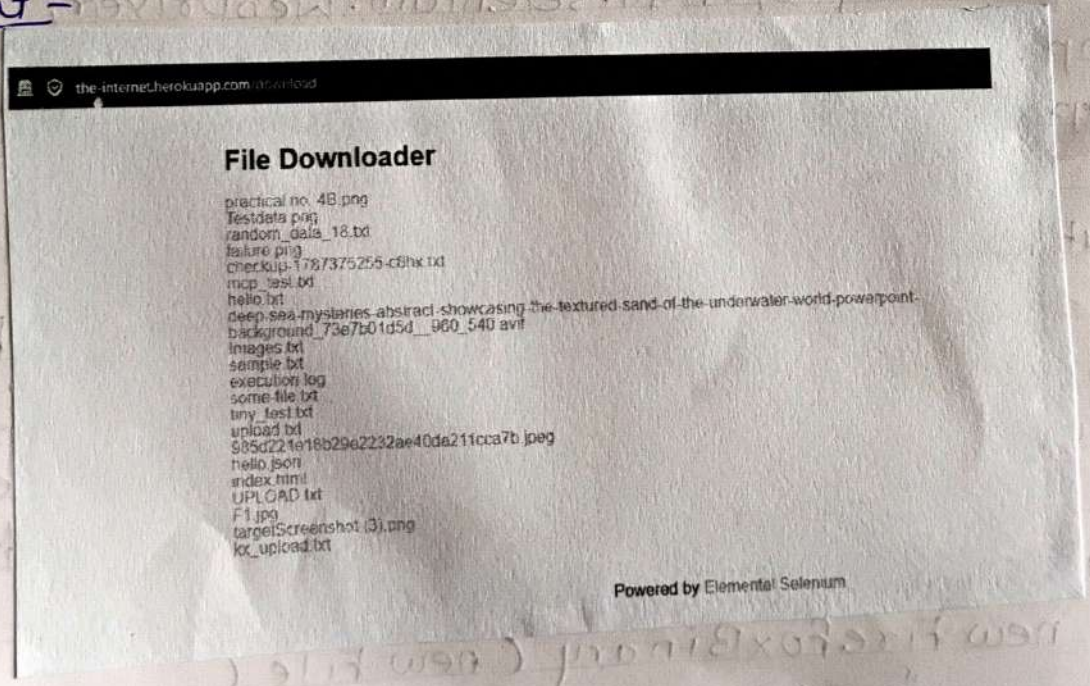
Teacher's Sign. :

step 4 -





output -



EXPERIMENT : No. 11. Practical No. 11

Page No. Date

Aim : Screenshot Capture and Logging Mechanism

Tool : Selenium WebDriver + log4j. implement - screenshot capture on test failure and integrate logging mechanism using log 4j. Generate execution logs and analyze error messages.

Steps -

1] Open Eclipse and create a project.

2] In Package Explorer, Find your project.

3] Expand the project and Find the src folder. Rightclick src → New → File → Name it. log4j.properties → Click Finish.

4] Eclipse will open the file, paste this :

```
log4j.rootLogger = INFO, file, console

log4j.appender.console = org.apache.log4j.ConsoleAppende
log4j.appender.console.layout=org.apache.log4j.PatternLayout
log4j.appender.console.layout.ConversionPattern = %d -
    %-5p %c - %m %n
log4j.appender.file =org.apache.log4j.FileAppender
log4j.appender.file.File= C:\\TestFiles\\ Execution.log
log4j.appender.file.layout=org.apache.log4j.PatternLayout
log4j.appender.file.layout.ConversionPattern = %d %-5p
%c - %m%n
```

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Teacher's Sign. : _____

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5] Save this file.
6] Download log4j 1.2.17 jar file and add it to your project's Build Path / Libraries.

7] Write the script :

— (screenshotLogTest.java) —

```
package pract11;
import java.io.File;
import org.apache.log4j.Logger;
import org.openqa.selenium.*;
import org.openqa.selenium.firefox.FirefoxDriver;

public class ScreenshotLogTest {
    static Logger log = Logger.getLogger(screenshot -
                                    LogTest.class);
    public static void main(String[] args) {
    System.setProperty("webdriver.gecko.driver",
            "C:\\Program Files\\Mozilla Firefox\\geckodriver.exe");
    WebDriver d = new FirefoxDriver();
    try {
     log.info("Test started");
     d.get("https://the-internet.herokuapp.com/login");
     log.info("Website opened");

     d.findElement(By.id("username")).sendKeys("tomsmith");
    d.findElement(By.id("password")).sendKeys("SuperSecret
                                    password!");
```

EXPERIMENT : No.

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Date

```
d.findElement(By.cssSelector("button[type='submit']")).
                                        click();
log.info("Login successful");
d.findElement(By.id("WrongElement"));
} catch(Exception e) {
  log.error("Test failed: "+e.getMessage());
  File src = ((TakesScreenshot)d)
     .getScreenshotAs(OutputType.FILE);
  File dest = new File("C:\\TestFiles\\Failure.png");
  src.renameTo(dest);
  log.info("Screenshot saved: C:\\TestFiles\\failure.png");
} finally {
   d.quit();
   log.info("Browser closed");
   log.info("Test completed");
  }
 }
}
```

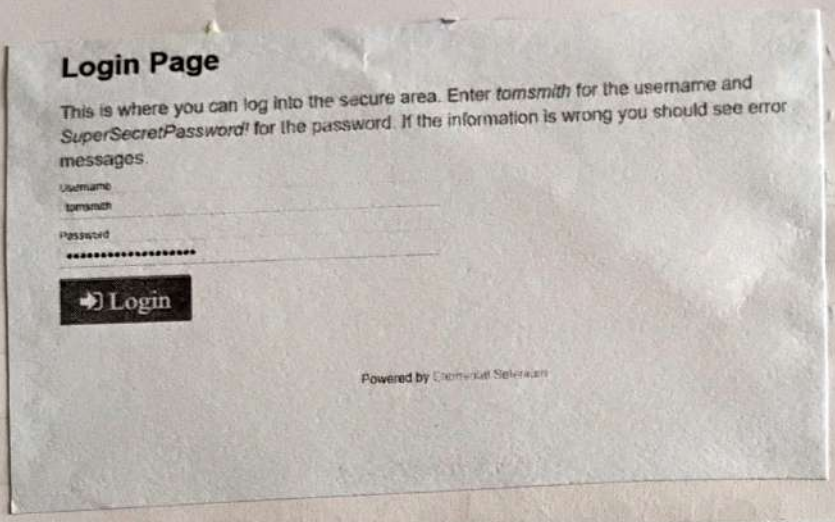
Step 8] Run this script from Eclipse IDE. After this's you will see login page.

9] Click on login button then you will see captured screenshot.

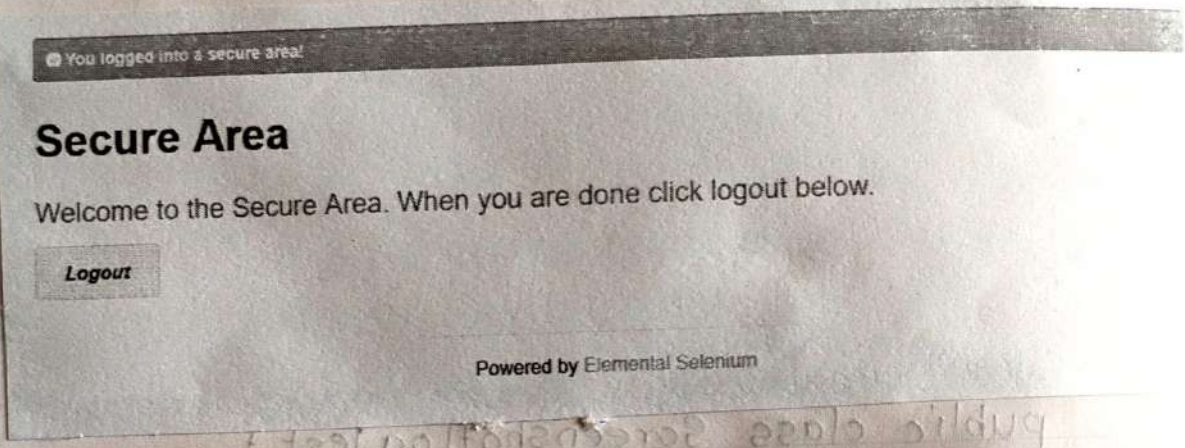
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step 9-



Captured Screenshot →




```
2026-08-22 11:58:06,627 INFO  pract10.ScreenshotLogTest - Screenshot saved: C:\TestFiles\failure.png
1787380086632   Marionette      INFO    Stopped listening on port 52445
JavaScript error: resource://gre/modules/PrivateBrowsingUtils.sys.mjs, line 50: TypeError: can't access property "Qu
Dynamically enable window occlusion 1
2026-08-22 11:58:07,170 INFO  pract10.ScreenshotLogTest - Browser closed
2026-08-22 11:58:07,170 INFO  pract10.ScreenshotLogTest - Test completed
```

EXPERIMENT : No. 12. Practical No. 12

Page No. Date

Aim : Web Element Interaction and synchronization
Handling Tool: Selenium WebDriver(Java)
Develop a script to automate interaction with various web elements (textbox, radio button, checkbox, drop-down, alert, and frames). Implement implicit and explicit waits for synchronization handling.

Steps-

1] Open Eclipse and create a java project.

2] Create package and class file in project and add jar files to it.

3] Write the script in java-

— (SeleniumDemo.java) —

```
package p12;
import java.util.concurrent.TimeUnit;
import org.openqa.selenium.*;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.support.AbstractFindBy-
                                        Builder;
import org.openqa.selenium.support.ui.*;

public class SeleniumDemo {
    public static void main(String [] args){
        System.setProperty("webdriver.gecko.driver",
    "C:\\Program Files\\Mozilla Firefox\\geckodriver.exe");
```

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```
WebDriver driver = new FirefoxDriver();
driver.manage().timeouts().implicitlyWait (
                10, TimeUnit.SECONDS);

driver.get ("https://demoqa.com/automation-practice-form");
WebDriverWait wait = new WebDriverWait(driver, 10);
WebElement textbox = wail.until(
 ExpectedConditions.visibilityOfElementLocated (By.id(
("username")));
 textbox.sendKeys ("Admin");

driver.findElement (By.id ("male")).click();
 driver.findElement (By.id ("terms")).click();
 Select dropdown =
        new Select(driver.findElement (By.id ("country")));
dropdown.deselectByVisibleText ("India");
driver.findElement(By.id ("alertButton")).click();

Alert alert = wait.until (
        ExpectedConditions.alertIsPresent ());
System.out.println(alert.getText());
alert.accept();
wail.untill (ExpectedConditions.frameToBeAvailableAnd-
                SwitchToIt(By.id ("myFrame")));

driver.findElement (By.id ("frameTextbox")).sendkeys
                ("Hello Frame");
driver.switchTo().defaultContent();
driver.quit(); }}
```

EXPERIMENT : No.

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Date

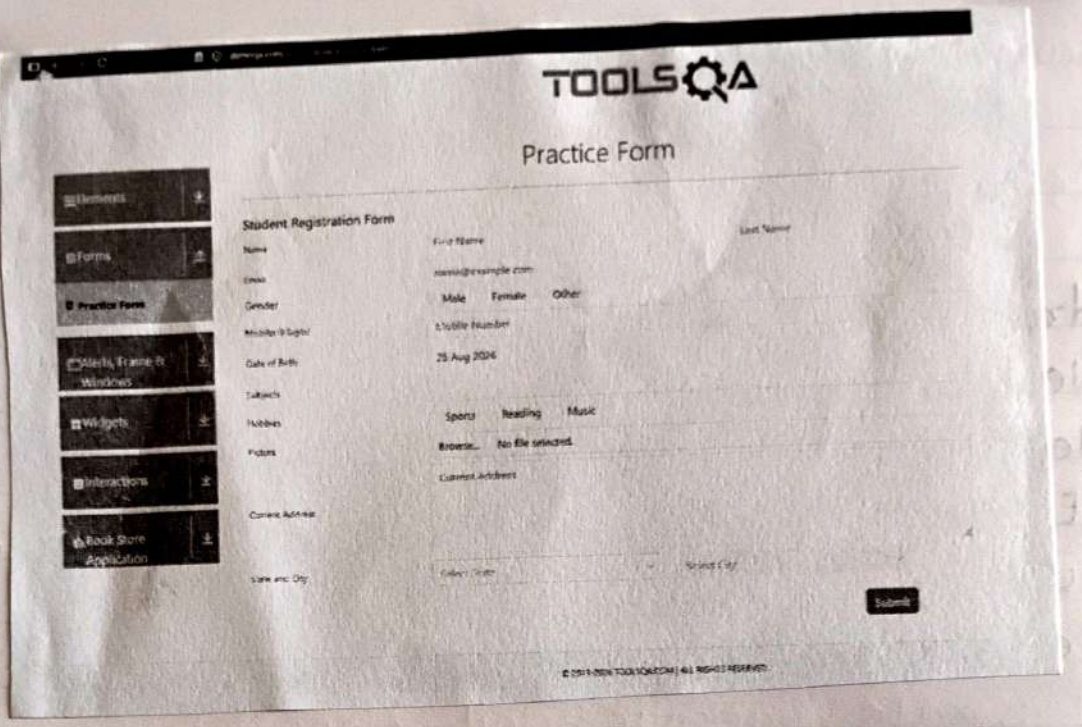
4] Run the script from Eclipse.

5] After running the script, you will see the practice form. Fill that form and click submit.

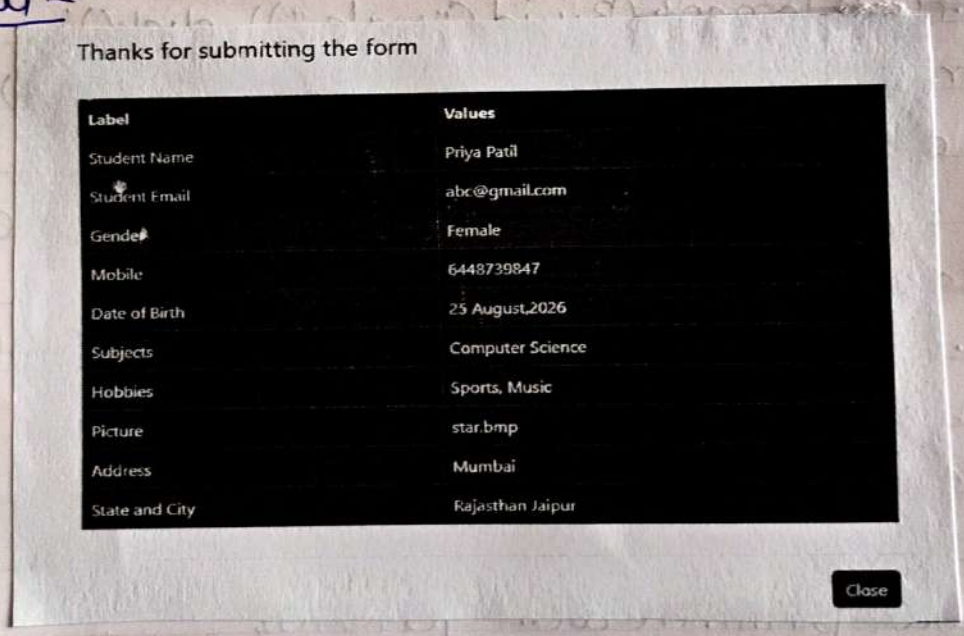
Sundaram

Teacher's Sign. :

5]


TOOLSQA
Practice Form
Elements
Forms
Practice Form
Alerts, Frame & Windows
Widgets
Interactions
Book Store Application
Student Registration Form
Name
Email
Gender
Mobile (10 Digits)
Date of Birth
Subjects
Hobbies
Picture
Current Address
State and City
First Name
Last Name
Male
Female
Other
25 Aug 2024
Sports
Reading
Music
Browse...
No file selected.
Submit

output -


Thanks for submitting the form
Label
Values
Student Name
Priya Patil
Student Email
abc@gmail.com
Gender
Female
Mobile
6448739847
Date of Birth
25 August,2026
Subjects
Computer Science
Hobbies
Sports, Music
Picture
star.bmp
Address
Mumbai
State and City
Rajasthan Jaipur
Close

EXPERIMENT : No. 13. Practical No. 13

Page No.
Date

Aim : Data Extraction and Store Analysis from Web Table
Tool : Selenium WebDriver. Automate extraction of student marks from a dynamic web table and determine the number of student scoring about at least one subject.

Pre-Requisites-
1) Check that you have Eclipse IDE.
2) check that you have JXL-JAR
3) check that you have the Excel file ("student.xls") that we'll be working on for reading data.

Steps-
1] Open Eclipse and create a java project.

2] Create a package and a class in project and add JXL jar file to it.

3] Write the script-

— (Excelreader.java) —

```
package excelread;
import java.io.File;
import java.io.IOException;
import jxl.Cell;
import jxl.CellType;
import jxl.Sheet;
```


Teacher's Sign. :

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Date

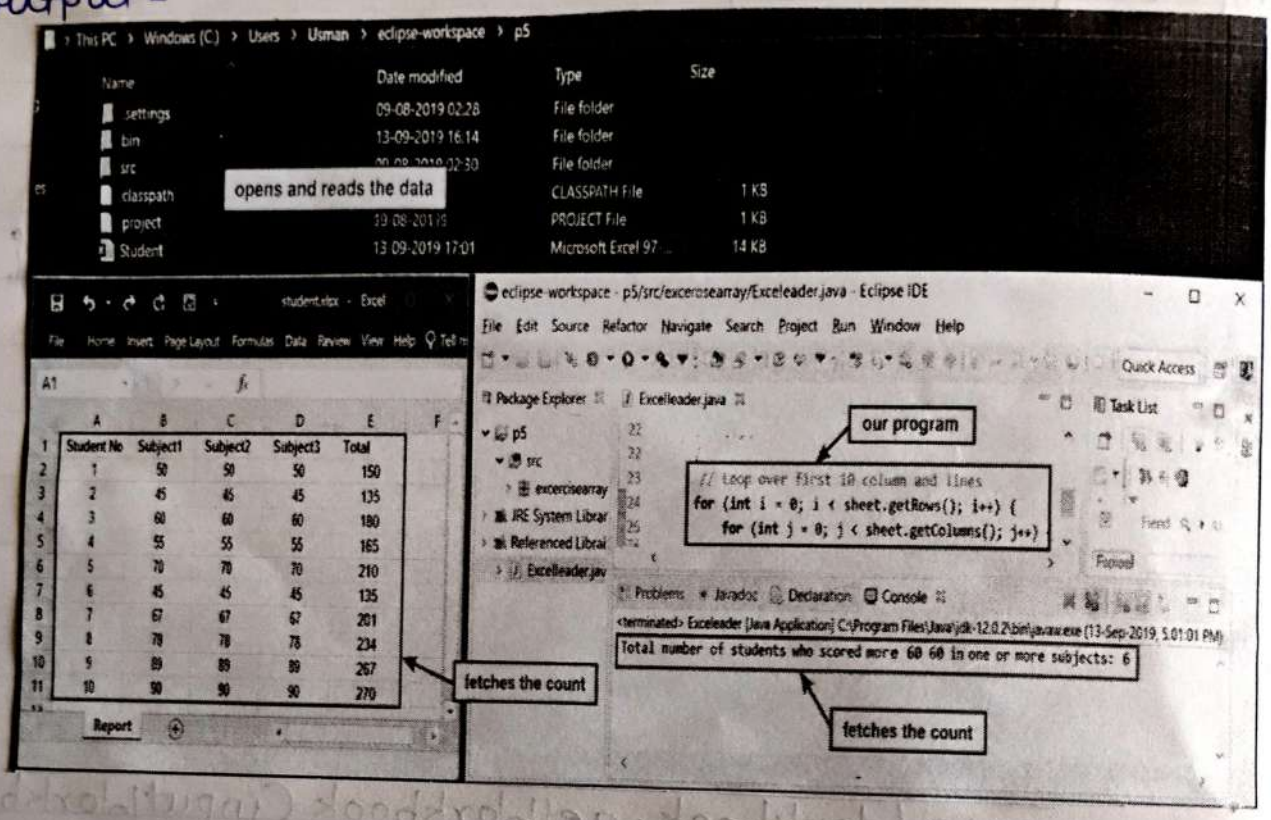
```
import jxl.Workbook;
import jxl.read.biff.BiffException;
  public class Excelreader {
private string inputFile;
  public void setInputFile (string inputFile) { this.
inputFile = inputFile; }
  public void read() throws IOException {
File inputWorkbook = new File (inputFile);
    Workbook w;
    boolean flag = false;
    int count = 0;
    try {
      w = Workbook.getWorkbook (inputWorkbook);
      sheet sheet = w.getSheet (0);
     for (int j = 0 ; j < sheet.getRows(); j++) {
     for (int i = 0; i < sheet.getColumns()-1; i++){
     Cell cell = sheet.getCell (i,j);
        if (cell.getType () == = CellType.NUMBER){
if(Integer.parseInt (cell.getContents()) > = 60 {
 flag = true; if (flag = = true) { count ++;
 flag = false;
}
 break;
  }
 }
 }
}
```

Teacher's Sign. :

```
System.out.println("Total number of students who
scored more than 60 in one or more subjects: "
+ count);
 }
 catch (BiffException e){e.printStackTrace();}
}
public static void main (String [] args) throws -
IOException {
 Excelreader test = new Excelreader();
test.setInputFile ("C:\\Users\\Usman\\eclipse work -
space\\ps\\student.xls");
   test.read();
 }
}
```

4] Run the file from Eclipse IDE.

output -


This PC › Windows (C:) › Users › Usman › eclipse-workspace › p5
Name
Date modified
Type
Size
settings
09-08-2019 02:28
File folder
bin
13-09-2019 16:14
File folder
src
File folder
classpath
CLASSPATH File
1 KB
project
PROJECT File
1 KB
Student
13-09-2019 17:01
Microsoft Excel 97 ...
14 KB
opens and reads the data
student.xlsx - Excel
File Home Insert Page Layout Formulas Data Review View Help
A1
Student No
Subject1
Subject2
Subject3
Total
1 50 50 50 150
2 45 45 45 135
3 60 60 60 180
4 55 55 55 165
5 70 70 70 210
6 45 45 45 135
7 67 67 67 201
8 78 78 78 234
9 89 89 89 267
10 90 90 90 270
Report
fetches the count
eclipse-workspace - p5/src/excercisearray/Exceleader.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Quick Access
Package Explorer
Excelleader.java
p5
src
excercisearray
JRE System Librar
Referenced Librai
Excelleader.jav
our program
// loop over first 10 column and lines
for (int i = 0; i < sheet.getRows(); i++) {
for (int j = 0; j < sheet.getColumns(); j++)
Problems
Javadoc
Declaration
Console
<terminated> Exceleader [Java Application] C:\Program Files\Java\jdk-12.0.2\bin\javaw.exe (13-Sep-2019, 5:01:01 PM)
Total number of students who scored more 60 60 in one or more subjects: 6
fetches the count

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Aim: Load Testing Using Apache JMeter

Tool: Apache JMeter

Design and execute load testing scenarios for a web application. Configure thread groups, ramp-up time, and loop count. Analyze response time, throughout and error percentage using graphical reports.

Pre-Requisites-

1) Download "jdk-12.0.2_windows-x64_bin.exe" from official website and install it.
2) To Download "Apache JMeter":
- Visit https://jmeter.apache.org/download_jmeter.cgi
- Under section "Apache JMeter 5.1.1 (Require Java 8+)"
- Under sub-section "Binaries", download "apache-jmeter-5.1.1.zip" file.
- Installation-
  - Extract the "apache-jmeter-5.1.1.zip" file.
  - Navigate to: apache-jmeter-5.1.1 > bin > the "ApacheJMeter.jar" file is your working space.

Steps-

1) Open "ApacheJMeter.jar" file and Name your Test plan.

2) Save your test plan. it will be saved as ".jmx" (i.e. Java Management Extensions).

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3] Create a Thread Group
(Right-click your test plan > Add > Thread (users) > Thread Group > Name your Thread Group as "myThread 1" > Save.)

4] Under Thread Group (myThread 1), add a "Sampler", namely HTTP Request :
(Right-click "myThread 1 > Add > Sampler > HTTP Request)

5] Now visit any webpage with a next path. Suppose we visit this website - lifecharger.org.when-in-doubt-do-this/
Then set website name as :-
- Server Name or IP = lifecharger.org &
- Path = when-in-doubt-do-this.

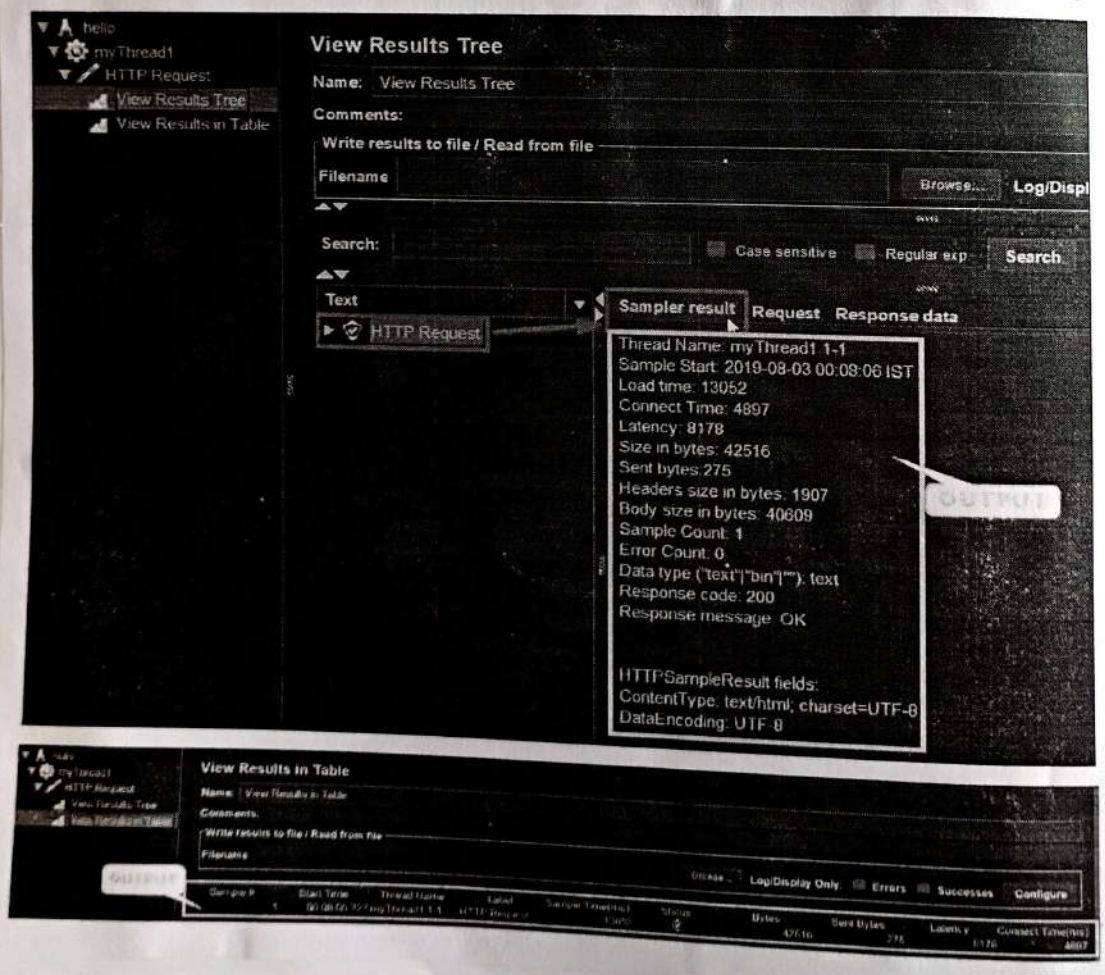
6] Under HTTP Request Sampler, add 2 "Listners" :-
1] Right-click on HTTP Request (sampler) > Add > Listener > View Results Tree &

2] Right-Click on HTTP Requst (sampler) > Add > listener > View Results in Table.

7] Save and Run the file and wait for a while. You'll see some Outputs.

Teacher's Sign. :

7] output -


hello
myThread1
HTTP Request
View Results Tree
View Results in Table
View Results Tree
Name: View Results Tree
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Displ
Search:
Case sensitive
Regular exp
Search
Text
HTTP Request
Sampler result
Request
Response data
Thread Name: myThread1 1-1
Sample Start: 2019-08-03 00:08:06 IST
Load time: 13052
Connect Time: 4897
Latency: 8178
Size in bytes: 42516
Sent bytes:275
Headers size in bytes: 1907
Body size in bytes: 40609
Sample Count: 1
Error Count: 0
Data type ("text"|"bin"|""): text
Response code: 200
Response message: OK
HTTPSampleResult fields:
ContentType: text/html; charset=UTF-8
DataEncoding: UTF-8
OUTPUT
View Results in Table
Log/Display Only:
Errors
Successes
Configure

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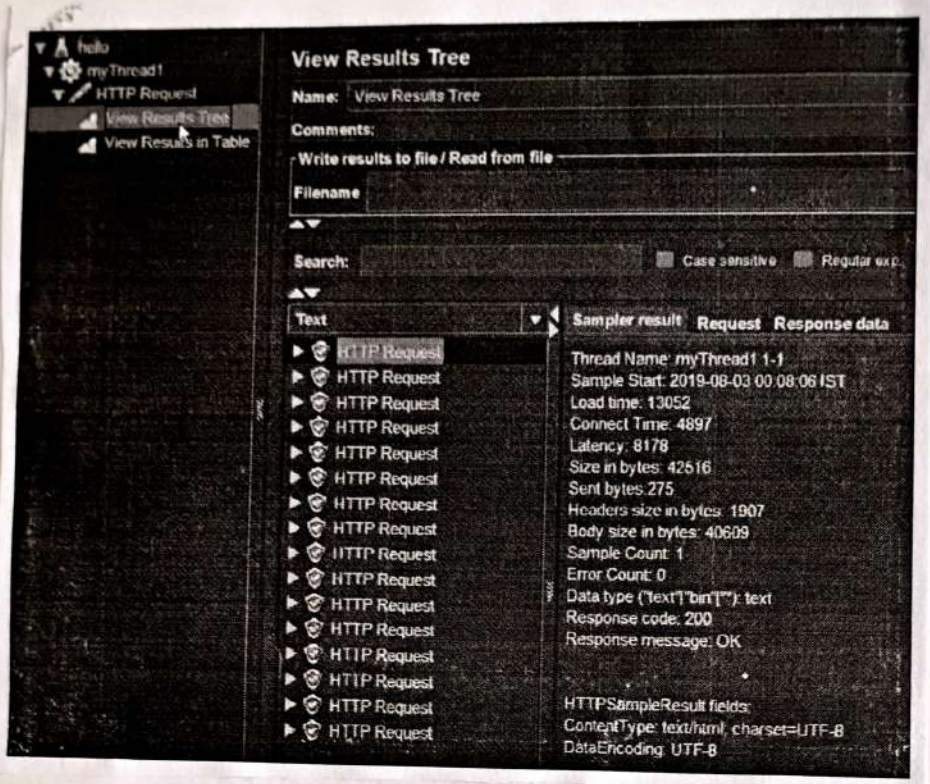
8] Now change some Thread Properties -
(users = 10 ; ramp-up period = 20 ; loop count = Forever)

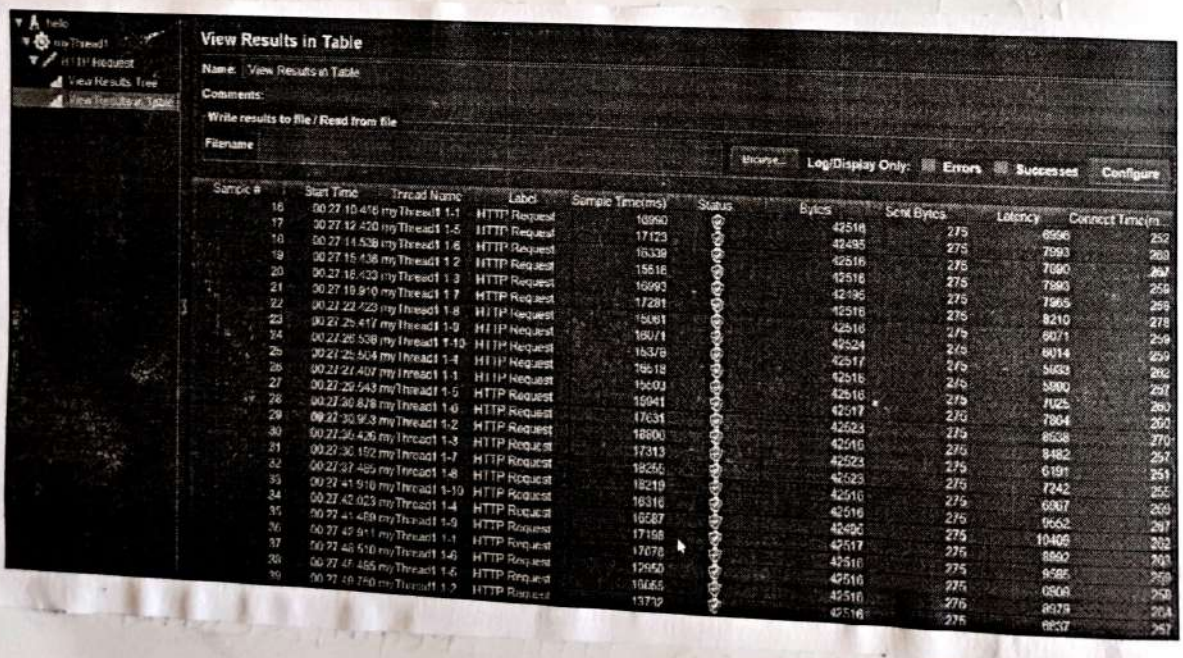
9) Save and Run the file and wait for a while.
you'll see some infinite outputs.

10) Finish!

Teacher's Sign. :

9] output -


hello
myThread1
HTTP Request
View Results Tree
View Results in Table
View Results Tree
Name: View Results Tree
Comments:
Write results to file / Read from file
Filename
Search:
Case sensitive
Regular exp.
Text
HTTP Request
Sampler result
Request
Response data
Thread Name: myThread1 1-1
Sample Start: 2019-08-03 00:08:06 IST
Load time: 13052
Connect Time: 4897
Latency: 8178
Size in bytes: 42516
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Body size in bytes: 40609
Sample Count: 1
Error Count: 0
Data type ("text"|"bin"|""): text
Response code: 200
Response message: OK
HTTPSampleResult fields:
ContentType: text/html; charset=UTF-8
DataEncoding: UTF-8


View Results in Table
Name: View Results in Table
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only:
Errors
Successes
Configure
Sample #
Start Time
Thread Name
Label
Sample Time(ms)
Status
Bytes
Sent Bytes
Latency
Connect Time(m...